GAUGE: Bridging the Command–Motion Gap in Black-Box Quadrupeds

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***Abstract*— Commercial and learned quadruped controllers often expose a planar velocity command while hiding the dynamics that map commands to realized motion. Payload, surface, controller, and estimator changes can therefore preserve balance yet distort navigation. We present GAUGE (Goal-Aware Uncertainty-Guided Exploration), a task-aware active calibration and shift-recovery layer for opaque quadruped velocity interfaces. A Bayesian command-to-motion model represents cross-axis coupling and asymmetric response, and a task-weighted integrated variance objective selects commands for a declared deployment distribution. Hard authorization, validation stopping, bounded inverse compensation, and residual-triggered recovery remain separate from acquisition. On 183 passive Unitree Go2 trials, a coupled affine map reduces held-out root-mean-square error (RMSE) by 54.5% relative to raw commands. Across controlled synthetic families, task weighting reaches a joint accuracy–uncertainty target in 18.67 trials, saving 39.5% over Latin hypercube sampling and 18.8% over D-optimal design. A candidate-noise oracle sensitivity changes the heteroscedastic result by one trial without changing the method ranking. Pinned Isaac Lab experiments show 9.3–31.7% held-out RMSE reduction and a matched-budget task-error advantage over D-optimal design on every navigation map. A paired residual-signature monitor produces 0/120 stationary-sequence alarms over 12,000 trials; across four shifts and 144 seeds per shift, detection is 99.3–100%, recovery is 97.9–100%, and active selection improves early recovery over passive updating. Failure-aware navigation is noninferior to a dense reference across six held-out maps. These results establish the active mechanism in controlled simulation and the need for a coupled response model in passive hardware data; qualitative real-Go2 trials further show reduced hesitation and contacts under fixed-planner compensation.**

I. Introduction

A quadruped can remain upright while executing the wrong motion. High-level controllers accept a desired planar velocity $\boldsymbol{u} = [v_x, v_y, \omega_z]^\top$, but the steady body velocity may contain gain error, dead zones, saturation, and cross-axis coupling. These errors are particularly consequential when a planner assumes that the exposed command is a calibrated control variable. Taouil *et al.* showed that learned motion models of a black-box quadruped controller can improve footstep planning [1]; related work has calibrated velocity-controlled kinematic models online for wheeled visual–inertial odometry [2]. The remaining question is how to collect the calibration trials themselves when hardware time is limited and the useful part of command space is task dependent.

Robot calibration has long treated measurement configuration as a design variable [3], [4]. Classical observability and D-optimal criteria prioritize parameter identification [5], [6]. Task-oriented calibration instead scores a design by uncertainty at the end-effector or another quantity of interest [7], [8]. We apply that principle to the external velocity interface of a fixed quadruped controller: trials are chosen for their reduction of predictive uncertainty over commands expected during deployment, not for uniform parameter-space coverage.

This layer differs from locomotion-policy adaptation. Rapid Motor Adaptation conditions a trained policy on latent environment estimates [9], residual models can update model-based legged controllers online [10], and recent embodiment adaptation identifies hardware changes from short interaction histories [11]. GAUGE neither retrains nor instruments the low-level controller. It identifies the mapping from user commands to measured steady motion, translates desired velocities through a constrained inverse, and reactivates calibration when predictive residuals indicate a shift.

This paper contributes (i) a compact Bayesian response model with coupled, asymmetric features and per-trial uncertainty; (ii) a conditional task-weighted integrated variance reduction (IVR) score with safe sequential design; (iii) an operational composition of authorization, validation stopping, constrained inversion, and latched shift recovery; and (iv) evidence spanning passive Go2 data, synthetic controls, fault injection, pinned physics, six held-out maps, and qualitative physical navigation. The novelty is the task-measure-to-command design and its closed-loop composition, not the individual regression, design update, or detector. Active selection and shift recovery are evaluated in controlled simulation.

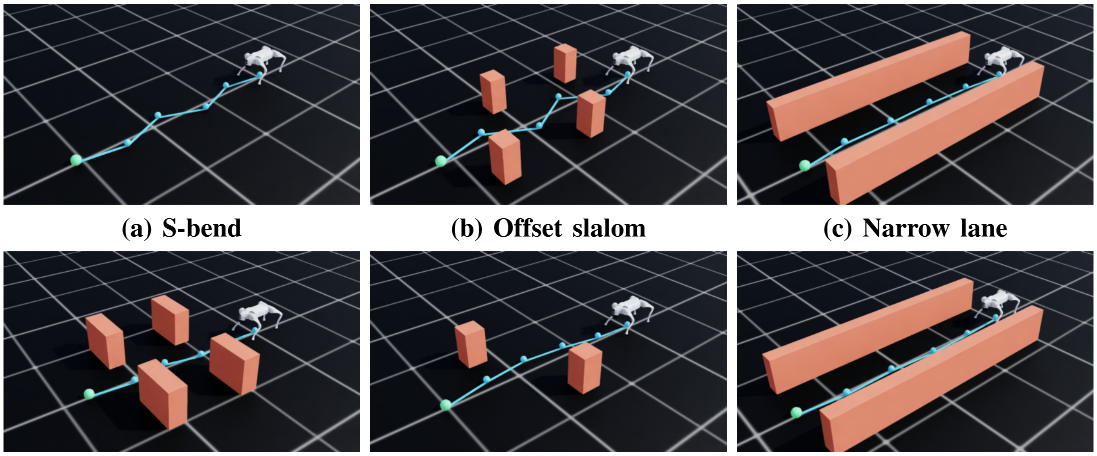
II. Related Work

A. Calibration and Experimental Design

Calibration accuracy depends on both model structure and measurement configuration [3], [4]. Observability and alphabetic designs optimize identification [5], [6]; task-oriented calibration instead targets downstream error [7], paralleling goal-oriented Bayesian design [8]. Equation (4) specializes a sequential Bayesian I/V-optimal criterion to safety-filtered velocity commands, integrates uncertainty over the planner-facing distribution, and sends the posterior to the same bounded inverse and change detector.

B. Legged Command Models and Adaptation

Simulation-trained policies achieve robust quadruped locomotion [12], [13], [14], but need not expose an exact velocity interface. Learned black-box motion models support planning [1], and dynamics or policy adaptation handles broader disturbances [15], [9], [11]. GAUGE instead calibrates a



(a) S-bend (b) Offset slalom (c) Narrow lane

(d) Double chicane (e) Weighted arc (f) Extended lane

Fig. 1. Elevated Isaac Sim views of the six held-out navigation maps. Cyan segments and spheres show the map waypoints; orange geometry shows the obstacles. The overlays are non-colliding visualization geometry and do not contribute to the quantitative endpoints.

fixed controller's external response, quantifies uncertainty, and compensates only inside a non-learned envelope.

III. Method

The low-level controller is fixed and opaque (Fig. 2). Each trial executes a constant body-frame command, extracts a robust steady-motion observation and covariance, and rejects invalid data before updating.

A. Bayesian Command-to-Motion Model

For command $\boldsymbol{u}$ and measured steady velocity $\boldsymbol{y} \in \mathbb{R}^3$, the coupled affine basis is $[1, v_x, v_y, \omega_z]$; the diagonal control zeros off-axis coefficients. The nonlinear 13-term basis adds pairwise products and $[v_k - \tau_k]_+, [-v_k - \tau_k]_+$, with $\boldsymbol{\tau} = [0.15, 0.10, 0.25]$. Non-intercept columns are standardized once on the safe pool. Hardware and navigation use the affine basis; synthetic and recovery studies use the nonlinear basis. Closed-loop priors regularize the linear projection toward identity.

For output axis a, the model is

$$y_a = \boldsymbol{\theta}_a^\top \boldsymbol{\phi}(\boldsymbol{u}) + \epsilon_a, \quad \epsilon_a \sim \mathcal{N}(0, \sigma_a^2 + r_a), \tag{1}$$

where r_a is the measurement-pipeline covariance diagonal and σ_a^2 is the frozen process variance. Given prior $\mathcal{N}(\boldsymbol{m}_{a0}, \boldsymbol{\Sigma}_{a0})$, sequential updates use

$$\boldsymbol{\Lambda}_a \leftarrow \boldsymbol{\Lambda}_a + \frac{\boldsymbol{\phi}\boldsymbol{\phi}^\top}{\sigma_a^2 + r_a}, \tag{2}$$

$$\boldsymbol{\eta}_a \leftarrow \boldsymbol{\eta}_a + \frac{\boldsymbol{\phi} y_a}{\sigma_a^2 + r_a}, \quad \boldsymbol{\Sigma}_a = \boldsymbol{\Lambda}_a^{-1}, \quad \boldsymbol{m}_a = \boldsymbol{\Sigma}_a \boldsymbol{\eta}_a. \tag{3}$$

The predictive mean is $\mu_a = \boldsymbol{m}_a^\top \boldsymbol{\phi}$; future-observation variance is $\boldsymbol{\phi}^\top \boldsymbol{\Sigma}_a \boldsymbol{\phi} + \sigma_a^2 + r_a$. Coverage uses the full variance and acquisition only its reducible epistemic term. Coupled input features represent cross-axis response while output posteriors remain independent. The error norm uses equal-axis unit normalization W; axiswise errors are also reported.

B. Task-Weighted Active Selection

A task distribution is represented by commands $\{\boldsymbol{g}_j\}_{j=1}^G$ and weights w_j within the forward-model domain. Nominal waypoint-planner outputs proxy the commands visited by inverse compensation. For candidate $\boldsymbol{c}$, a Sherman–Morrison update gives the conditional one-step IVR:

$$\mathcal{I}(\boldsymbol{c}) = \sum_{a=1}^{3} \sum_{j=1}^{G} w_j \frac{\left(\boldsymbol{\phi}(\boldsymbol{g}_j)^\top \boldsymbol{\Sigma}_a \boldsymbol{\phi}(\boldsymbol{c})\right)^2}{\sigma_a^2 + \tilde{r}_a(\boldsymbol{c}) + \boldsymbol{\phi}(\boldsymbol{c})^\top \boldsymbol{\Sigma}_a \boldsymbol{\phi}(\boldsymbol{c})}. \tag{4}$$

The default $\tilde{r}_a = 0$ because future covariance is unavailable, while Eq. (3) uses realized r_a; an oracle sensitivity supplies known synthetic noise. The planner maximizes $\mathcal{I}(\boldsymbol{c}) - \lambda_r R(\boldsymbol{c}) - \lambda_d D(\boldsymbol{c})$, where R and D penalize command magnitude and travel, over a finite pool. The hard filter acts only after ranking. Greedy batches use covariance-only fantasy updates; the sample-efficiency study sets both penalties to zero.

C. Safety, Stopping, Compensation, and Shift Recovery

The hard filter checks state validity, localization, battery, attitude, base height, command and slew bounds, coupled load, and workspace occupancy. It fails closed if no ranked candidate remains; an independent 50 Hz monitor can zero commands. Statistical value therefore cannot relax an execution limit.

Stopping requires minimum coverage and three consecutive checks of validation RMSE and integrated uncertainty; hard budgets take priority. Synthetic oracle crossings remain hidden evaluation endpoints. The inverse searches the bounded candidate set and minimizes

$$\|W(\boldsymbol{\mu}(\boldsymbol{u}) - \boldsymbol{v}^*)\|_2^2 + \lambda \|W(\boldsymbol{u} - \boldsymbol{v}^*)\|_2^2 + \rho \sum_a s_a^2(\boldsymbol{u}), \tag{5}$$

then passes the selected command through the hard filter.

Shift monitoring cycles through four safe commands $(-.20, 0, -.20)$, $(-.20, 0, .20)$, $(.12, -.10, -.15)$, and $(.12, .10, .15)$. Commissioning freezes the corresponding

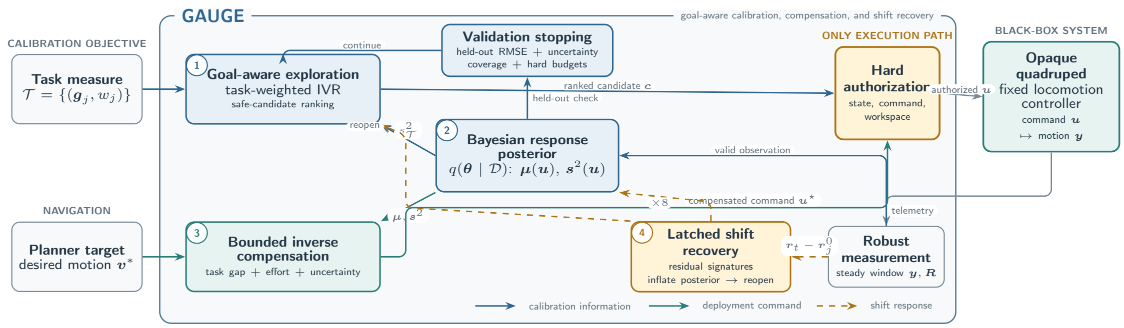


Fig. 2. GAUGE architecture. Task-weighted acquisition ranks calibration commands, but the hard filter alone authorizes execution. Valid observations update the Bayesian model; held-out validation and uncertainty control stopping. A latched residual alarm inflates posterior uncertainty and opens a bounded active-recovery phase. The inverse compensator uses the same bounded command set during navigation.

residual signatures $\boldsymbol{r}_j^0 = \boldsymbol{y}_j - \boldsymbol{\mu}(\boldsymbol{u}_j)$. For a later response to signature command $j(t)$, the detector evaluates

$$d_t = \left\| \operatorname{diag}(.14, .08, .18)^{-1} \left(\boldsymbol{r}_t - \boldsymbol{r}^0_{j(t)} \right) \right\|_2. \qquad (6)$$

An alarm requires $d_t > .30$ on at least two of the last four valid monitor trials and a two-trial minimum dwell, so an isolated response change cannot trigger recovery. The alarm latches, inflates epistemic covariance by a factor of eight without changing the posterior mean, and opens a 12-command recovery budget. Normal validation stopping resumes after recovery.

IV. Experimental Evaluation

A. Passive Hardware and Sample Efficiency

The hardware dataset contains 183 valid passive Go2 trials in three 61-trial sessions, referenced by globally localized Livox MID360 FAST-LIO. Leave-one- session-out folds compare raw, diagonal, and coupled linear/affine models. Sampled fits use 30 fixed-LHS commands from 122 training trials; dense fits use all 122. Each command has 0.6 s ramp-in, 0.8 s settling, 2.0 s measurement, and 0.6 s ramp-out.

Sample efficiency uses 20 paired seeds in affine, dead-zone, and heteroscedastic synthetic families. All methods share a six-command seed and 160-trial horizon over $[-1, 1] \times [-.5, .5] \times [-1.5, 1.5]$, with $\|(v_x, v_y)\|_2 \leq 1$. Frozen draws vary diagonal gain, off-axis coupling, bias, saturation, and noise. The task mixture centers on forward motion and left/right turns, with weights $(.50, .25, .25)$. The joint target is RMSE ≤ 0.04 and integrated epistemic variance ≤ 0.0015. Random, LHS, Sobol, D-optimal, no-task IVR, and a dense oracle are controls. A hidden 400-command grid determines crossing and a disjoint 1,024-command grid audits it. The primary one-sided paired Wilcoxon test compares trials against LHS after averaging families within seed; all baseline contrasts undergo Holm correction. Paired bootstrap intervals use 5,000 resamples for trials saved and 4,000 for fixed-budget errors.

Distribution sensitivity freezes acquisition, then reweights the three task components to forward-, left-, or right-heavy mixtures and finally expands to the uniform safe envelope; no command is reselected.

B. Closed-Loop Simulation and Safety Evaluation

Physics experiments use Isaac Lab v2.3.2, Isaac Sim 5.1.0, and PhysX [16]. Four contexts each use 20 paired seeds, 12 calibration commands, and eight held-out commands. Hardware and simulation share a Huber constant-twist estimator from body-frame pose increments; timestamp, constancy, sample-count, and steady-window checks reject invalid observations.

Safety tests replay 60 acquisition traces, 300 hazardous proposals, 20 safe controls, and 160 state faults under a 50 Hz monitor. Shift recovery uses two disjoint 72-seed blocks for each of four changes: gain/coupling; friction/load/gain; physical/nonlinear; and load/COM/gain. Fixed-model, passive-update, and GAUGE arms share 12 initial commands, four signatures, five monitor commands, a 12-command recovery budget, and rolling validation; early RMSE averages trials 4–9. Stationary monitoring spans 30 sequences per context and 100 trials per sequence (12,000 trials). A 30-seed selector ablation holds recovery fixed while replacing task IVR with no-task IVR, D-optimal, LHS, or random selection.

C. Held-Out Navigation

Navigation crosses six disjoint maps (Fig. 1), seven calibration methods, and 72 paired seeds per map (3,024 episodes). All arms share the controller, 50 Hz interlock, 10 Hz waypoint follower, geometry, and randomization. GAUGE and matched controls use 12 calibration commands; the dense reference uses 30, followed by eight common validation commands. Task IVR ranks a 512-command safe pool under the planner-facing command measure.

Routes are 2.85–3.60 m with a minimum 0.79 m opening and 0.20 m collision radius. The command envelope is $[-.40, .40] \times [-.30, .30] \times [-.70, .70]$, with linear norm at most 0.45 m/s. Success reaches the 0.25 m goal ball without collision or serious event by 60 s; failures retain the 60-s cap. Calibration takes 31.2 s for 12 trials and 78.0 s for 30 (52.0 and 98.8 s with validation). Marginal rates use exact intervals; paired endpoints use 4,000 seed resamples. Noninferiority margins are a 1.25 capped-time ratio and -0.05 success difference.

For physical navigation, a Unitree Go2 runs the same fixed

DRL-DCLP local planner [17] with either direct velocity commands or the frozen GAUGE inverse between planner and robot. Two static-obstacle scenes and one crossing-pedestrian scene are each repeated five times per condition; matched-time frames, source video, and exact extraction times are retained.

V. Results

A. Model Need and Calibration Efficiency

Figure 5a shows a consistent held-out benefit from coupled calibration on the passive Go2 data. Pooled RMSE falls from 0.06605 for raw commands to 0.04563 for diagonal-affine and 0.03009 for coupled-affine calibration. The nested diagonal-linear and coupled-linear fits obtain 0.04828 and 0.03517, respectively. Coupled linear therefore outperforms diagonal affine without an intercept, isolating cross-axis response. Coupled-affine calibration reduces error by 54.45% versus raw commands and 34.07% versus diagonal-affine calibration. Its three held-out-session values are 0.02888, 0.03158, and 0.02974, and its 30-trial fit is within 3.33% of the 122-trial dense reference. The gain is concentrated laterally, where RMSE falls from 0.08459 to 0.02569 and the yaw-to-lateral coefficient remains 0.119–0.129 across holdouts. These data establish passive hardware model need rather than active hardware efficiency.

The synthetic benchmark isolates the acquisition rule (Fig. 5b). GAUGE reaches the joint target in 18.67 trials, versus 30.87 for LHS, 30.67 for random, 25.72 for Sobol, 23.00 for D-optimal, and 25.67 without task weighting. The respective savings are 39.52%, 39.13%, 27.41%, 18.84%, and 27.27%; all paired tests remain significant after Holm correction. The primary LHS contrast saves 12.20 trials [10.73, 13.73]. Horizon-end active RMSE is 0.00846, and future-observation 95% predictive coverage is 94.14%.

Realized task error shows the same ordering (Fig. 5d). At budgets 18, 24, and 30, task-IVR RMSE is 0.02143, 0.01684, and 0.01420. Its paired improvement is 0.00392–0.00466 over D-optimal and 0.00545–0.00881 over no-task IVR; all six 95% intervals exclude zero.

Frozen designs remain effective under navigation-family reweighting (Fig. 5e): forward- and right-heavy tests retain positive intervals against every control, while the left-heavy D-optimal contrast becomes positive by 30 trials. A uniform safe-envelope measure reverses the ordering at 24 trials; task-IVR error is 3.21 times its declared-task error (Fig. 5f). Deployment support expansion therefore requires an updated task measure or a general-purpose design.

The uncertainty gate binds in all 60 active conditions. Task IVR requires 18, 18, and 20 trials across the three families, versus 19, 19, and 31 for D-optimal and 23, 23, and 31 without task weights (Fig. 5c). A disjoint 1,024-command audit gives crossing RMSE 0.01796–0.02315 and axiswise 95% coverage 0.917–0.955. Task IVR also saves at least 3.33 trials over D-optimal across nine predeclared gate pairs. Supplying exact heteroscedastic candidate noise changes its crossing from 20 to 21 trials while preserving the 11-trial advantage over both principal controls.

In closed-loop simulation (Fig. 3e), raw-to-calibrated RMSE is 0.15199 to 0.10635 for affine distortion, 0.15054 to 0.10282 for dead-zone distortion, 0.10741 to 0.09742 for low friction with payload and COM offset, and 0.09554 to 0.07849 on rough terrain with payload and COM offset. Relative reductions span 9.30–31.70%. The corresponding paired absolute-improvement intervals are [0.04290, 0.04780], [0.04495, 0.05002], [0.00981, 0.01017], and [0.01477, 0.01889]. All 20 paired seeds improve in each scenario.

B. Stopping, Safety, and Shift Recovery

The stopping test records no premature stop across 60 trajectories; median and p95 overshoot are two trials. The hard filter rejects 300/300 declared hazards, 0/20 safe controls, and all 160 sampled-state faults. In physics, the 50 Hz monitor's maximum abort interval is 20 ms. These software and simulator checks are not physical safety certification.

The stationary-monitor experiment produces 0/120 sequence alarms over 12,000 valid trials; the two-sided exact 95% upper bound is 0.0303. Under shift, detection is 143/144 for the combined friction, load, and gain change and the gain-and-coupling change, and 144/144 for the physical-and-nonlinear change and the load, COM, and gain change. The lowest exact 95% detection-rate bound is [0.9619, 0.9998]. GAUGE recovers in 141/144 friction-plus-payload trials and 144/144 for the other shifts; its lowest exact interval is [0.9403, 0.9957]. Worst p95 detection and recovery delays are 3.9 and 9 trials, or 10.1 s of monitor-command time and 46.8 s for adaptation-plus-validation cycles under the fixed schedule.

GAUGE improves early RMSE over passive updating by 0.00936 for the combined friction, load, and gain change, 0.01053 for the gain-and-coupling change, 0.01847 for the physical-and-nonlinear change, and 0.01192 for the load, COM, and gain change (Fig. 4f). Their paired 95% intervals are [0.00811, 0.01065], [0.00961, 0.01140], [0.01774, 0.01917], and [0.01092, 0.01291]. Its terminal RMSE ranges from 0.09762 to 0.11227; every upper interval endpoint is below 0.116 and the declared 0.14 absolute gate (Fig. 4g). These estimates pool both independent blocks and retain the fixed-model, passive-update, and GAUGE pairing within every seed.

The selector ablation isolates acquisition (Fig. 4h). Task IVR lowers early RMSE by 0.02192 [0.02046, 0.02343] versus D-optimal, 0.01583 [0.01418, 0.01761] versus no-task IVR, 0.00937 [0.00655, 0.01212] versus LHS, and 0.00986 [0.00654, 0.01309] versus random selection. With no serious event, this establishes that early recovery depends on task-directed commands rather than detection and updating alone.

C. Navigation Consequence

GAUGE succeeds in 72/72 episodes on five maps and 70/72 on weighted arc, while raw commands succeed in 0/72 or 1/72 (Fig. 6a). Every GAUGE collision count is zero. The failure-aware 60-s capped-time gain over raw has 95%

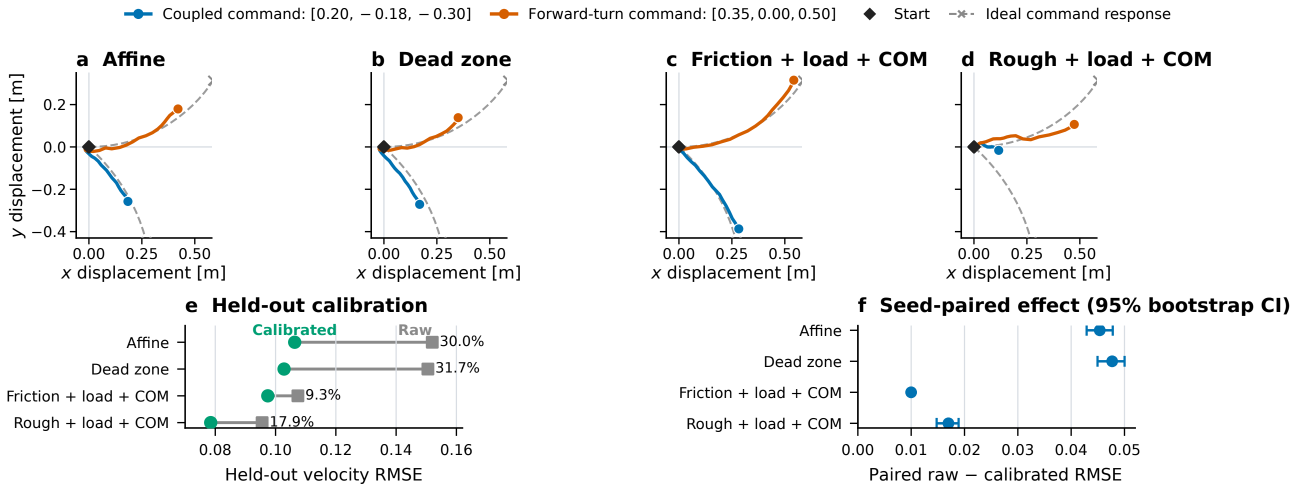


Fig. 3. Closed-loop simulation responses and calibration. (a)–(d) Body-displacement trajectories for coupled and forward-turn validation commands in a representative paired run; colored curves are measured, dashed gray curves and crosses are the ideal response to the declared command/timing, and diamonds mark the common start. (e) Raw and calibrated held-out velocity RMSE across 20 paired seeds; labels give relative reduction. (f) Seed-paired absolute improvement with 95% 4,000-sample bootstrap intervals. The trajectories are qualitative examples; all inference uses paired summaries specified before evaluation.

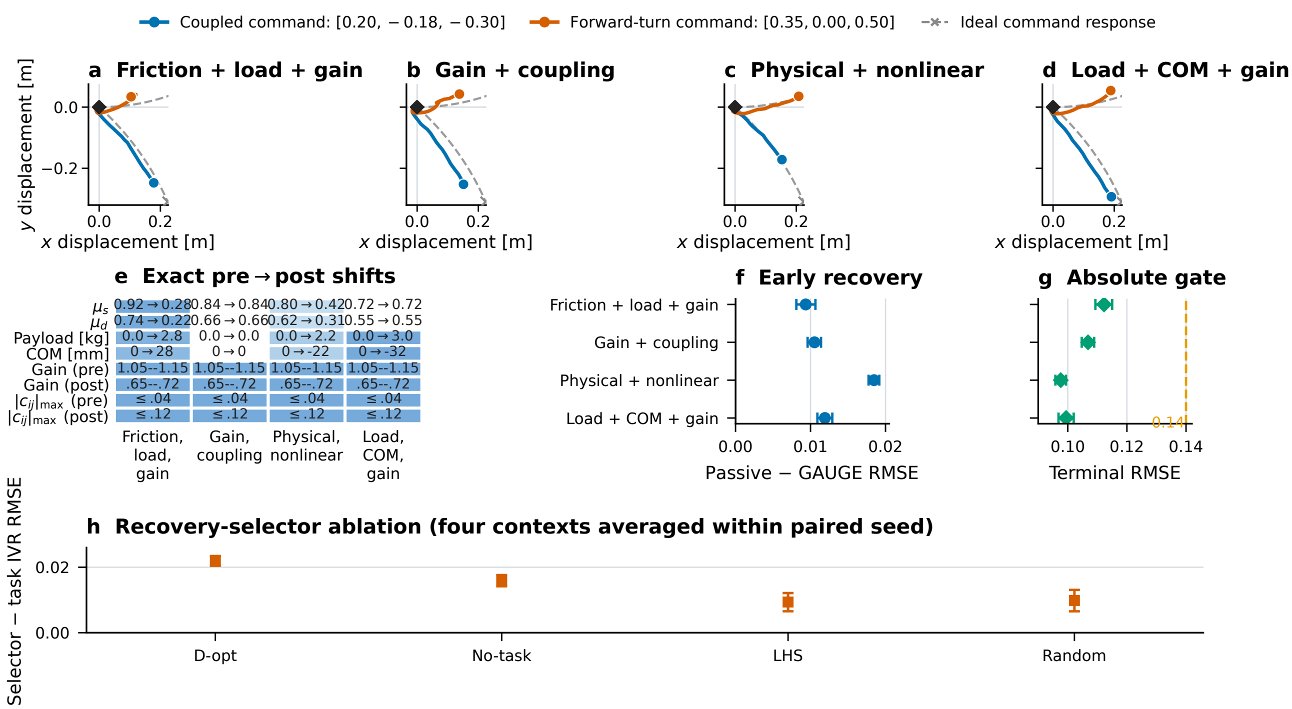


Fig. 4. Held-out simulator shifts and active recovery. (a)–(d) Measured body-displacement trajectories and dashed ideal command responses for the same validation commands in a representative paired run. (e) Exact pre→post static and dynamic friction, payload, fore–aft COM, per-axis gain range, and maximum off-axis coupling. Blue cells identify altered quantities, and every cell gives the exact pre- or post-shift value. The mixed condition additionally introduces dead zones, saturation, lag, and observation noise. (f) GAUGE improvement over passive updating in the pre-specified early window. (g) GAUGE terminal RMSE and the declared 0.14 gate. (h) Recovery-selector ablation, averaging the four contexts within each of 30 paired seeds; positive values favor task IVR. Error bars are paired 95% 4,000-sample bootstrap intervals. Recovery inference pools two disjoint 72-seed blocks per shift; stationary false alarms are evaluated separately over 120 sequences and 12,000 trials.

lower endpoints of 23.45–32.68 s (Fig. 6b). At the same 12-trial budget, task IVR reduces command-validation RMSE over D-optimal by 9.98–12.28% on every map (Fig. 6c); its advantage over no-task IVR is positive on five maps and unresolved on S-bend.

The worst GAUGE/dense capped-time ratio upper bound is 1.0736, below the 1.25 margin, and every paired success-difference lower bound exceeds −0.05 (Fig. 6d). The two weighted-arc timeouts are neither collisions nor serious events. Navigation excludes calibration: GAUGE executes

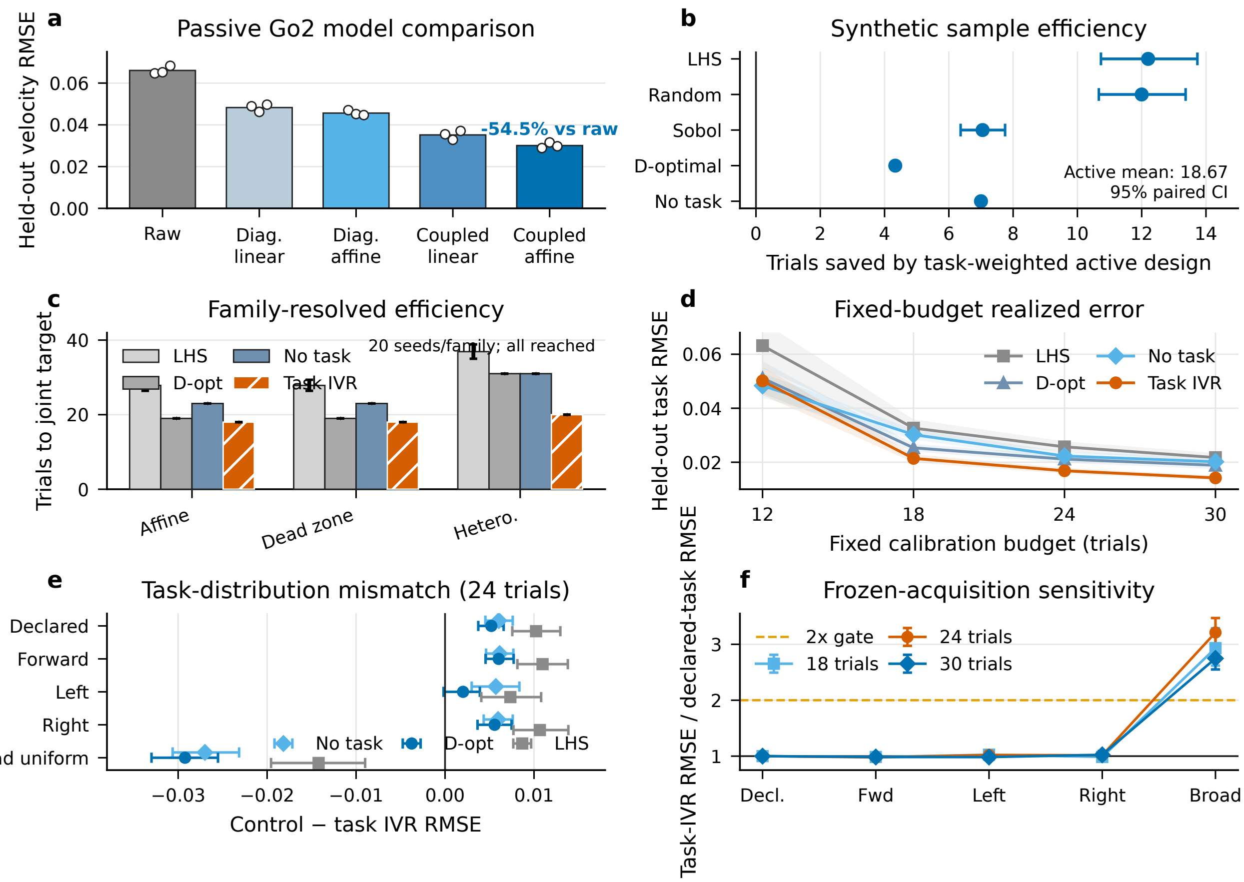


Fig. 5. Calibration evidence. (a) Passive real-Go2 held-out RMSE for raw, diagonal-linear, diagonal-affine, coupled-linear, and coupled-affine commands. (b) Paired trials saved by task IVR; bars are 95% seed-bootstrap intervals. (c) Family-resolved time to the same joint target; every method reaches it in all 20 seeds per family. (d) Realized held-out task RMSE at matched trial budgets; bands are 95% paired-seed bootstrap intervals. (e) Frozen-acquisition performance under task-distribution mismatch at 24 trials; positive control-minus-task-IVR values favor task IVR. (f) Task-IVR error relative to the declared measure; the uniform safe-envelope negative control crosses the pre-specified $2\times$ sensitivity gate.

31.2 s of acquisition commands versus 78.0 s for dense, or 52.0 versus 98.8 s after common validation commands are included. Across 432 episodes it records 10 non-serious safety events and no serious event. Identical seeds, policy, planner, interlock, maps, and navigation logic isolate calibration in this simulator pipeline.

D. Physical Navigation Consequence

At equal elapsed times, direct commands produce visible stop-and-correct motion and contact at static obstacles, while GAUGE advances with less hesitation and contact (Fig. 7a–b). Compensation also clears the pedestrian crossing earlier (Fig. 7c). Across five repetitions per condition, every scene improves in hesitation, contact, or both. The fixed planner isolates the effect of planner-facing command compensation.

VI. Discussion and Limitations

The experiments separate model need, task-directed acquisition, and downstream navigation. No-task and D-optimal controls isolate task weighting; noise and gate sensitivities preserve the ordering. Reweighting within the navigation family retains the benefit, whereas safe-envelope support expansion does not, so the task measure must change with deployment support. Dense and matched-budget navigation controls establish downstream noninferiority, while the raw arm exposes the cost of an uncalibrated interface.

The joint crossing is uncertainty-limited in all 60 active conditions, so the measured efficiency is faster acquisition of posterior confidence. Independent 400- and 1,024-command grids audit accuracy and predictive coverage; trace replay separately verifies stopping logic. Shift evaluation likewise separates four-command commissioning, 12,000 stationary monitor trials, shifted detection, and the fixed early-recovery window.

Hardware evidence comprises passive calibration and qualitative closed-loop navigation on one Go2; active selection and shift recovery use a pinned simulator. The three physical scenes and five repetitions support behavior-level comparison, not trajectory-level effect sizes. The compact basis targets steady response rather than arbitrary history dependence, gait switching, or rapid transients. Logged standardized routes and physical interface, load, center-of-mass, and friction shifts are required for quantitative hardware effect sizes and shift-transfer claims.

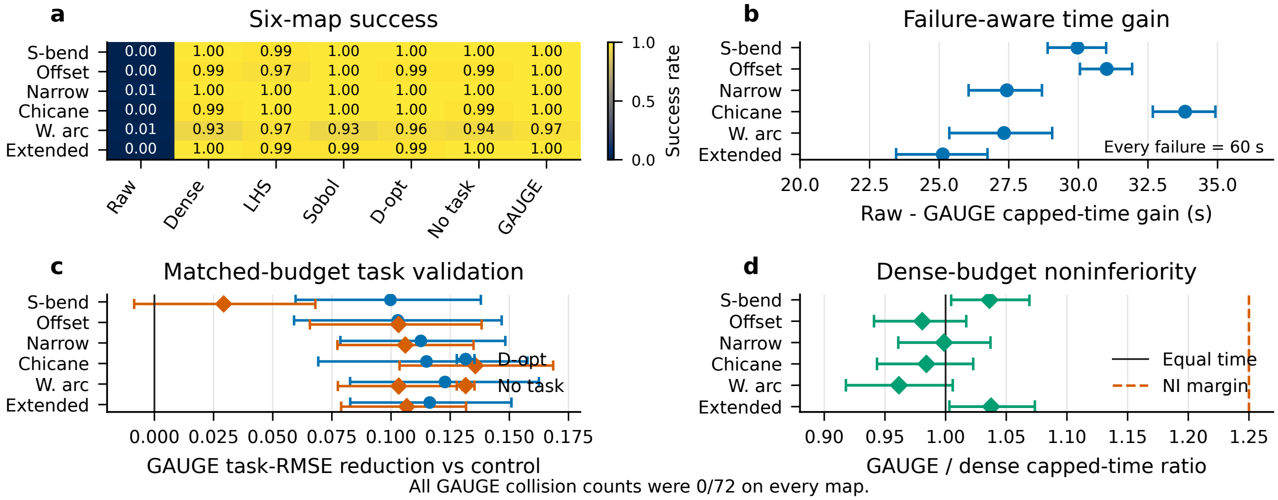


Fig. 6. Held-out simulator navigation (72 seeds/cell). (a) Success rate. (b) Paired raw-minus-GAUGE gain in the 60-s capped-time estimand, which retains every failure as 60 s. (c) Matched-budget validation-RMSE reduction from task IVR relative to D-optimal and no-task IVR; bars are paired 95% intervals. (d) GAUGE/dense capped-time ratios and the declared 1.25 noninferiority margin. Every GAUGE collision count is 0/72.

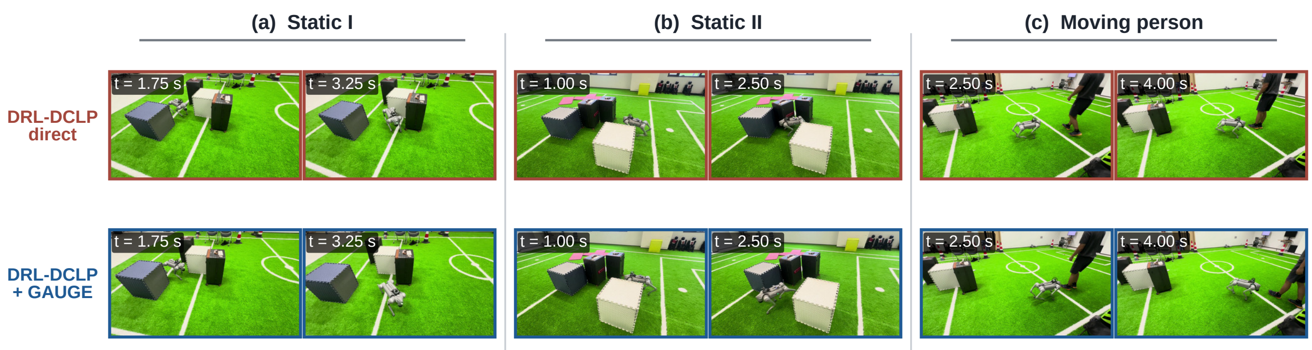


Fig. 7. Matched-time real-Go2 navigation with a fixed DRL-DCLP planner. Rows compare direct commands (red) and GAUGE-compensated commands (blue) in (a)–(b) two static-obstacle scenes and (c) a crossing-pedestrian scene. Each condition has five repetitions; the displayed sequences are representative, not independent statistical samples.

VII. Conclusion

GAUGE directs calibration trials toward task-relevant predictive uncertainty and composes coupled Bayesian regression with hard safety, validated stopping, bounded inversion, and latched recovery. Passive Go2 data establish model need; controlled synthetic and Isaac Lab experiments establish sample efficiency, early recovery, and downstream navigation. Real-Go2 sequences demonstrate the planner-facing effect of command compensation with DRL-DCLP; online active acquisition and physical shift recovery remain to be evaluated.

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